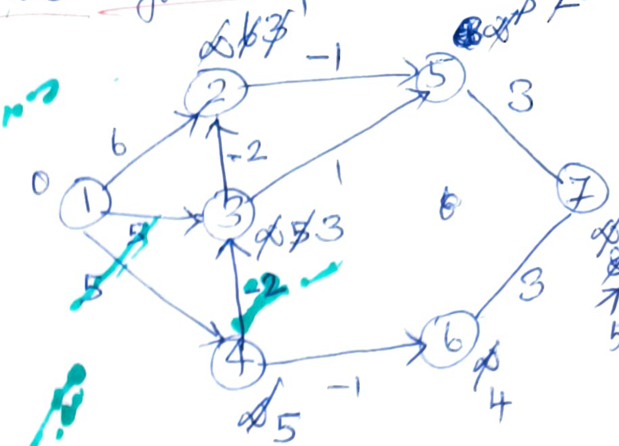
Single Source · Shortest Path

Bellman Ford Algorithm



Starting vertex: (1)

Dijkstra
— won't work perfectly for negative edges.

B.F follows Dynamic Programming Strategy

Relax for n−1 vertices
$7-1=6$.

Select all edges starting from anywhere.

List of edges — (1,2)(1,3)(1,4) (2,5)(3,2)(3,5)(4,3)(4,6) (5,7) and (6,7)

Relaxation:

between any pair of vertices (u,v) if

$d[u] + c(u,v) < d[v]$)

$d[v] = d[u] + c(u,v)$

(1) start vertex = 0 (starting vertex)
others as — ∞

Iteration 1

$(1,2) = 0+6 < \infty \Rightarrow 6$

$(1,3) - 0+5 < \infty \Rightarrow 5$

$(1,4) = 0+5 < \infty \Rightarrow 5$

$(2,5) = 6+(-1) < \infty = 5$

$(3,2) = 5-2 < 6 = 3$

$(3,5) = 5+1 > 6$ — keep as it is

$(4,3) = 5-2 < 5 = 3$

$(4,6) = 5-1 < \infty = 4$

$(5,7) = 5+3 < \infty = 8$

$(6,7) = 4+3 < 8 = 7$

Iteration 2

$(1,2) = 0+6 \Rightarrow 3$ keep

$(1,3) = 0+5 > 3$ keep

$(1,4) = 0+5 = 5$ keep

$(2,5) = 3-1 < 5 \Rightarrow 2$

$(3,2) = 3-2 < 3 = 1$

$(3,5) = 3+1 > 2$ - keep

$(4,3) = 5-2 = 3$ - keep

$(4,6) = 5-1 = 4$ keep.

$(5,7) = 2+3 < 7 = 5$

$(6,7) = 4+3 > 5$ keep

Iteration 3

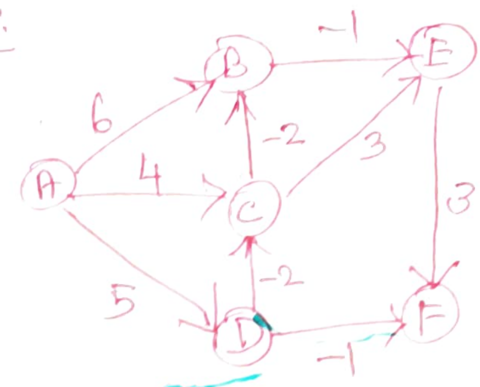
(1,2) = 1
(1,3) = 3
(1,4) = 5
(2,5) = 1 − 1 = 0
(3,2) = 1 Keep
(3,5) = 0 Keep
(4,3) = 3 Keep
(4,6) = 4
(5,7) = 0+3 < 5 = 3
(6,7) = 4+3 > 3 = keep

Iteration 4

(1,2) = 1
(1,3) = 3
(1,4) = 5
(2,5) = 0
(3,2) = 1
(3,5) = 0
(4,3) = 3
(4,6) = 4
(5,7) = 3
(6,7) = 3

No change has occurred
∴ stop relaxation process.

Ex:



∴ The shortest Path

1 − 0
2 − 1
3 − 3
4 − 5
5 − 0
6 − 4
7 − 3

Time Complexity of Bellman ford.

$O(|E| \cdot |V| - 1)$

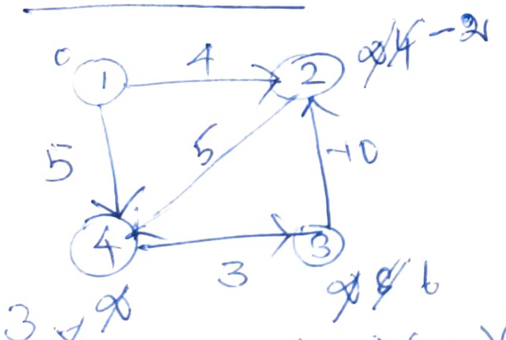
$O(n \cdot n)$

$O(n^2)$

for complete Graph we have $\frac{n(n-1)}{2}$ edges

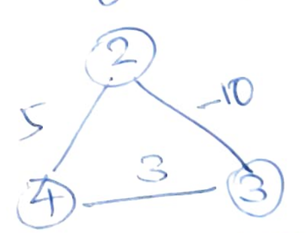
$\frac{n(n-1)}{2} \cdot (n-1)$ $O(n^3)$

Drawback



edges (3,2), (4,3) (1,4) (1,2) (2,4)

There is a cycle of edges



5+3−10 = −2

Total weight of the cycle is −ve. So every time

If there is a −ve weight cycle then Graph cannot be solved.

∴ Bellman ford fails to solve if there is a −ve weight cycle but it can detect the −ve edge cycle